

Benchmarking and Performance Analysis of LLM Inferencing Systems

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Bachelor's Thesis Project
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Workload Setup



- **Goal:** Evaluate inference performance of LLMs under short queries with large shared context.
- **Model:** Qwen 0.5B, Tiny Llama 1.1B, served using vLLM/SGLang/Transformers.
- **Hardware:** NVIDIA RTX6000 48GB GPU.
- **Metrics:** Throughput (req/s), latency, GPU utilization.

Parameter	Value/Range
System Prompt Length	≈1500 tokens (shared across requests)
Query Rate	5–700 req/sec
Query Length	10–20 tokens
Output Length	30–50 tokens
Batch Size	Adaptive

Table: Workload Characteristics

Experiment Setup: Engine Comparison



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- Compare inference engines: **Transformers**, **vLLM**, **SGLang**, **NEO**.
- Batch of 24 requests, each sharing the same system prompt.
- Each request: $\sim$20 input tokens $\rightarrow$ $\sim$40 generated tokens.
- Measure **total time taken** to process the batch.

Hypothesis

Engines leveraging prefix caching (e.g., SGLang) should perform far better than naive batching in Transformers.

Batch of requests through various Engines

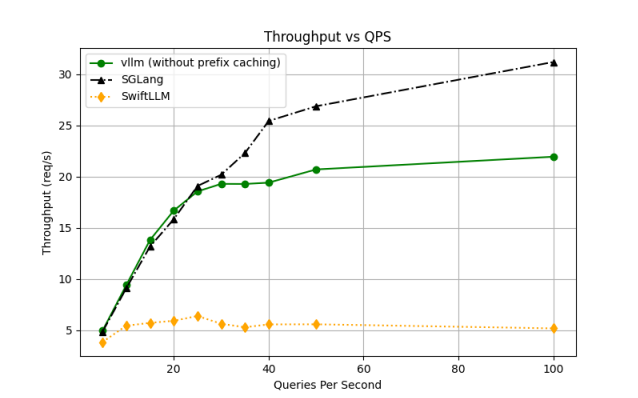


Figure: Throughput comparison across varying workloads

- **SwiftLLM:** lightweight vLLM clone (∼2000 LOC), but doesn't performs that well.
- **SGLang:** achieves best performance via radix-tree prefix sharing.

Technique Used	Time (s)
Transformers (batch = 1)	30.21
Transformers (batch = 2)	19.51
Transformers (batch = 3)	18.64
vLLM (no system prompt pre-filling)	0.45
SGLang (fork)	1.03
SGLang (run_batch)	0.40
NEO	0.55

Table: Measuring the time taken to process a batch of 24 requests using different techniques

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Experiment Setup: Request Timelines



- Aim: Measure time spent in **prefill** vs **decode** stages for 20 req/sec workload (sending a new request after every 0.05 seconds).
- Using vLLM V0 engine without chunked prefill.
- Tool: Custom logging of per-request timestamps.

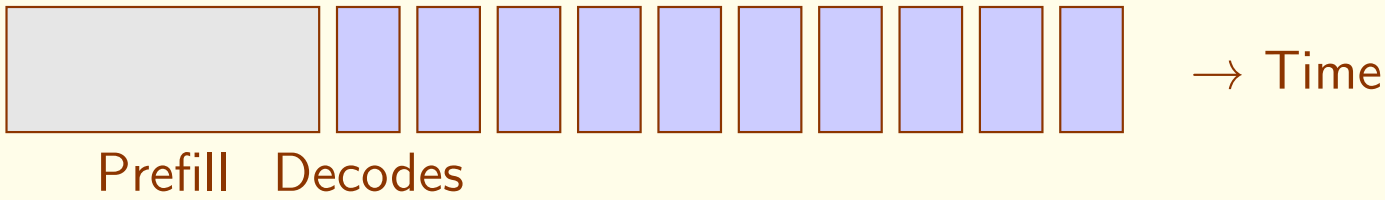


Figure: Simplified per-request timeline

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Results: LLM Request Timelines

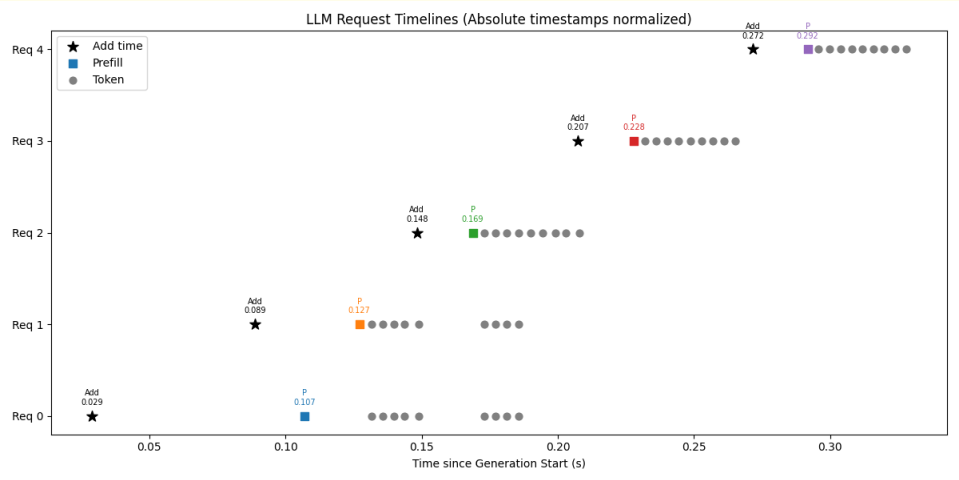


Figure: 20 Requests/sec – No chunked prefill – vLLM V0

- Prefill dominates runtime due to large prompt.
- Prefill and decode cannot overlap here.
- Low TTFT but high total latency.

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Results: Throughput vs Load for vLLM

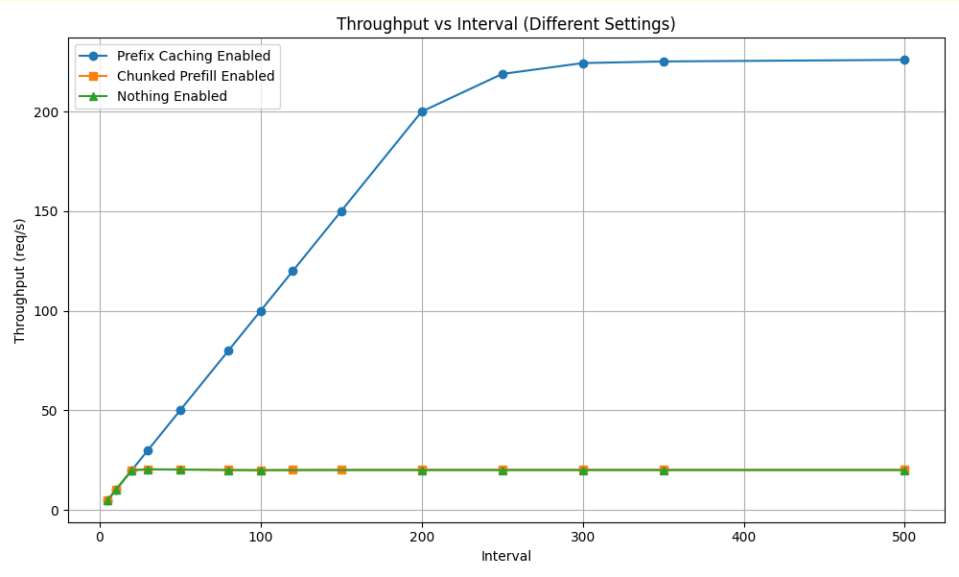


Figure: Throughput vs Load

- Varying the input load, and comparing different modes.
- Chunked prefill: limited benefit for small queries.
- Prefix caching: major boost to throughput.
- Batch tuning further improves efficiency.

Observation

Saturation begins at ∼300 req/sec — compute bottleneck.

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Results: NSYS Measurements

Goal

Understand compute vs memory utilization and bottlenecks.



Figure: NSYS Metrics Snapshot

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Observation: NSYS Profiling



- Compute: SM Active and Warp occupancy.
- Memory: vLLM books full GPU memory upfront.
- vLLM scheduler stats show the actual KV Cache Usage.
- PCIe Bandwidth low $\rightarrow$ small workloads, minimal CPU transfer.

GPU Memory Breakdown

Component	Memory Occupied
Weights	0.93 GiB
Peak Activation	4.07 GiB
Non-Torch	0.02 GiB
CUDA Graph	0.41 GiB
KV Cache: Intersection (Prefix)	1,280 tokens / 0.0157 GiB
KV Cache: Extras for Requests	640 tokens (10 requests × 64 tokens) / 0.0078 GiB
KV Cache: Free	2,672,880 tokens / 32.8445 GiB

Table: GPU Memory Breakdown and KV Cache Usage

Observation: Memory Profiling



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- KV Cache footprint is tiny — prefix reuse highly memory-efficient.
- Most GPU memory remains free $\rightarrow$ headroom for larger batches.
- Suggests potential for multi-family prefix caching in future work.

Generation of these families

- Let's say we already have a large system prompt of length 1500 tokens.
- We generate multiple families of requests by prepending a unique family identifier to the system prompt.
- For example, for family 1, we prepend "AA", for family 2, "AB", and so on.
- This ensures that although the families have a large substring in common, but their prefixes always differ, and the KV Cache block hashes won't match.
- This in turn implies, that we need to compute and store separate KV Cache blocks for each family.

Experiments with Multiple Families

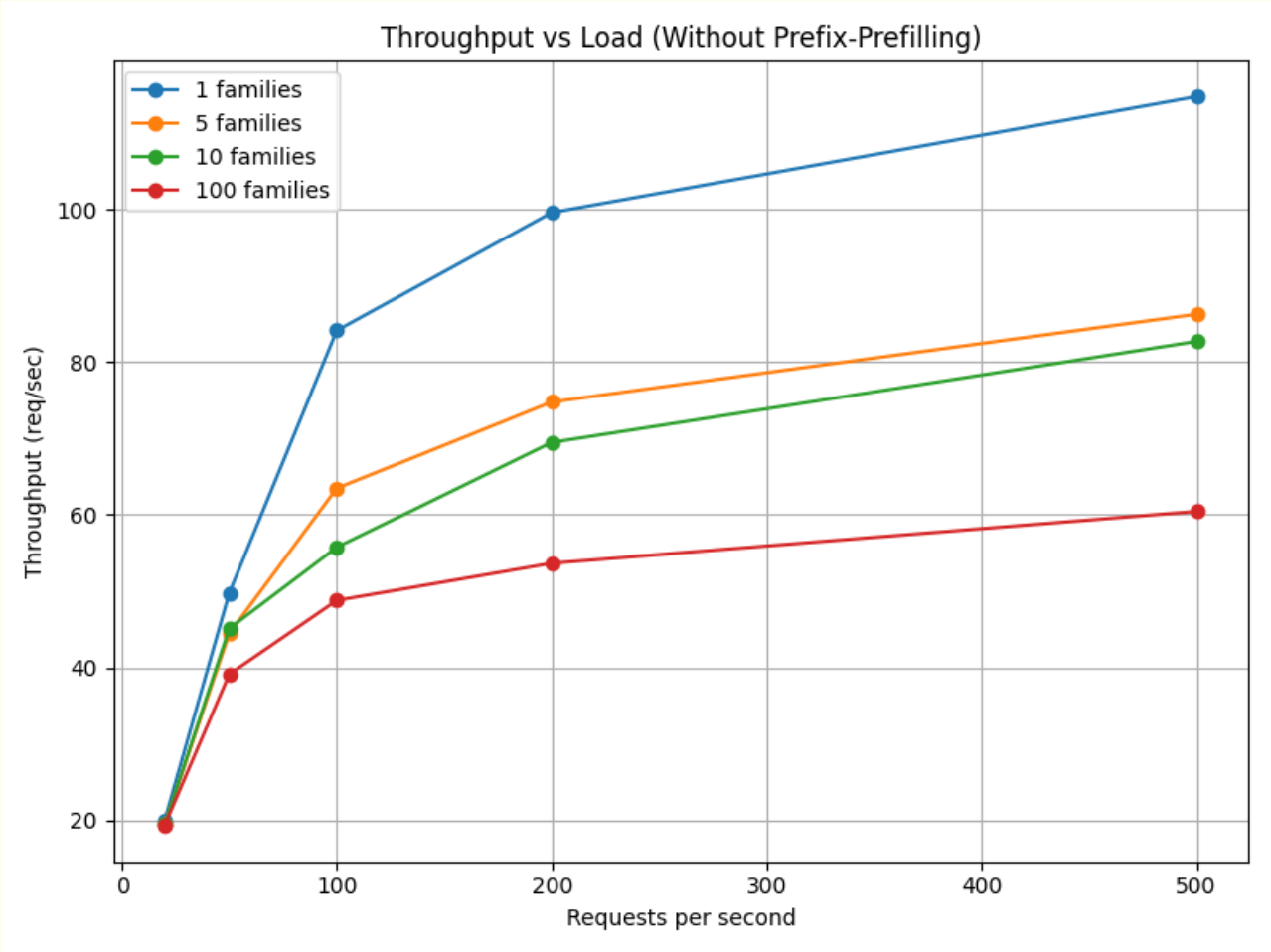


Figure: Throughput vs Load without prefix pre-filling

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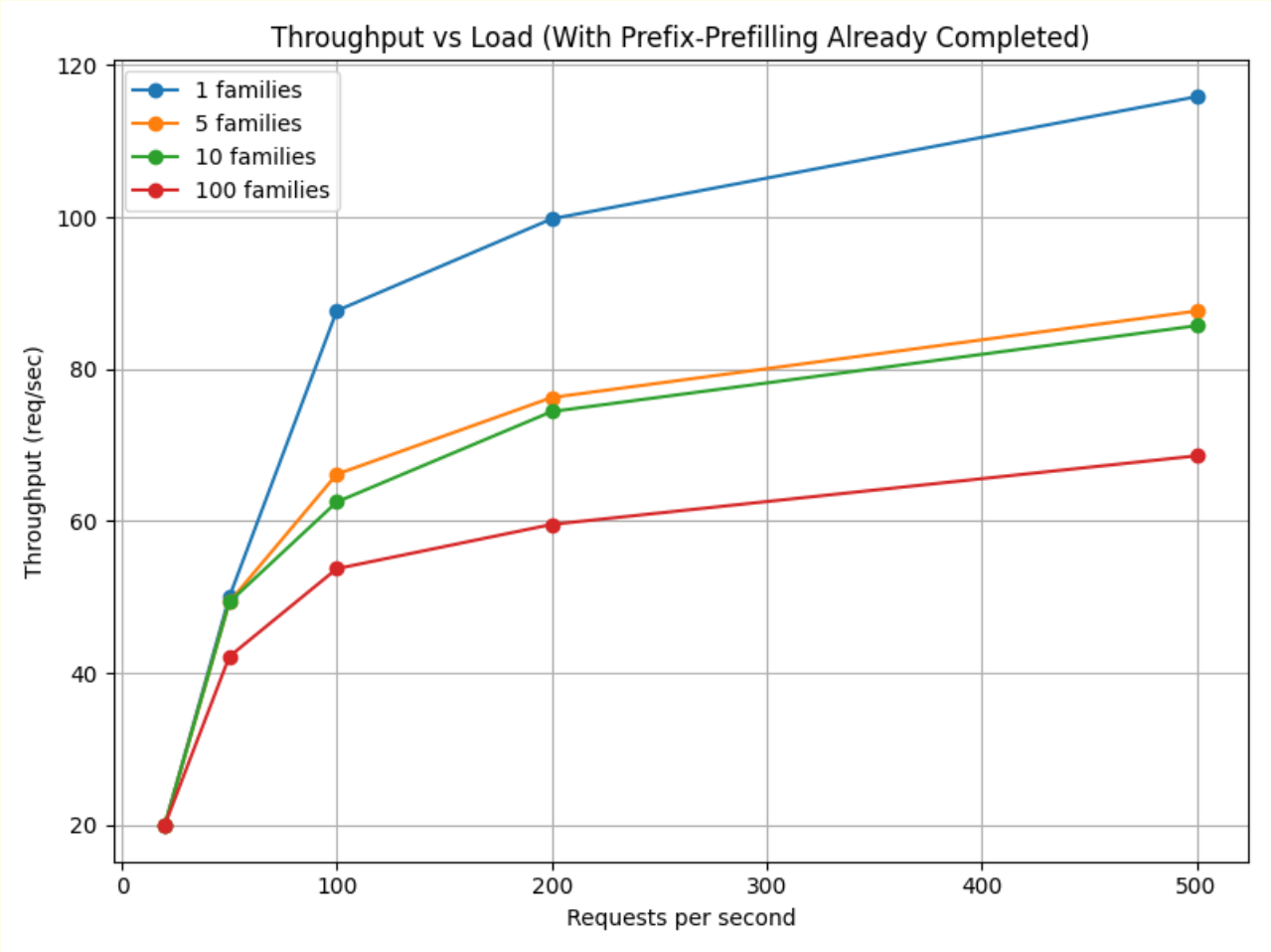


Figure: Throughput vs Load with prefix pre-filling

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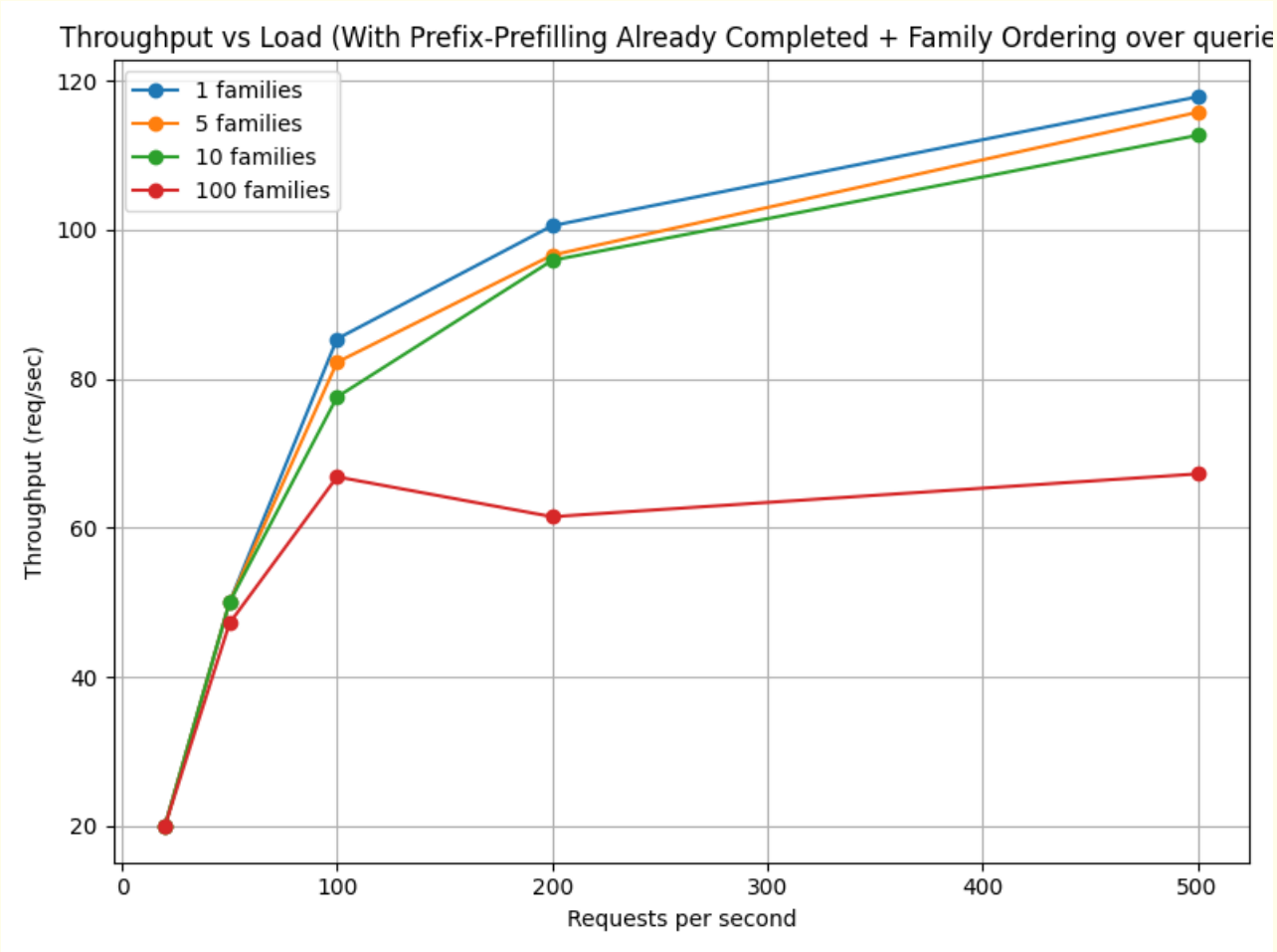


Figure: Throughput vs Load with prefix pre-filling with ordered family IDs

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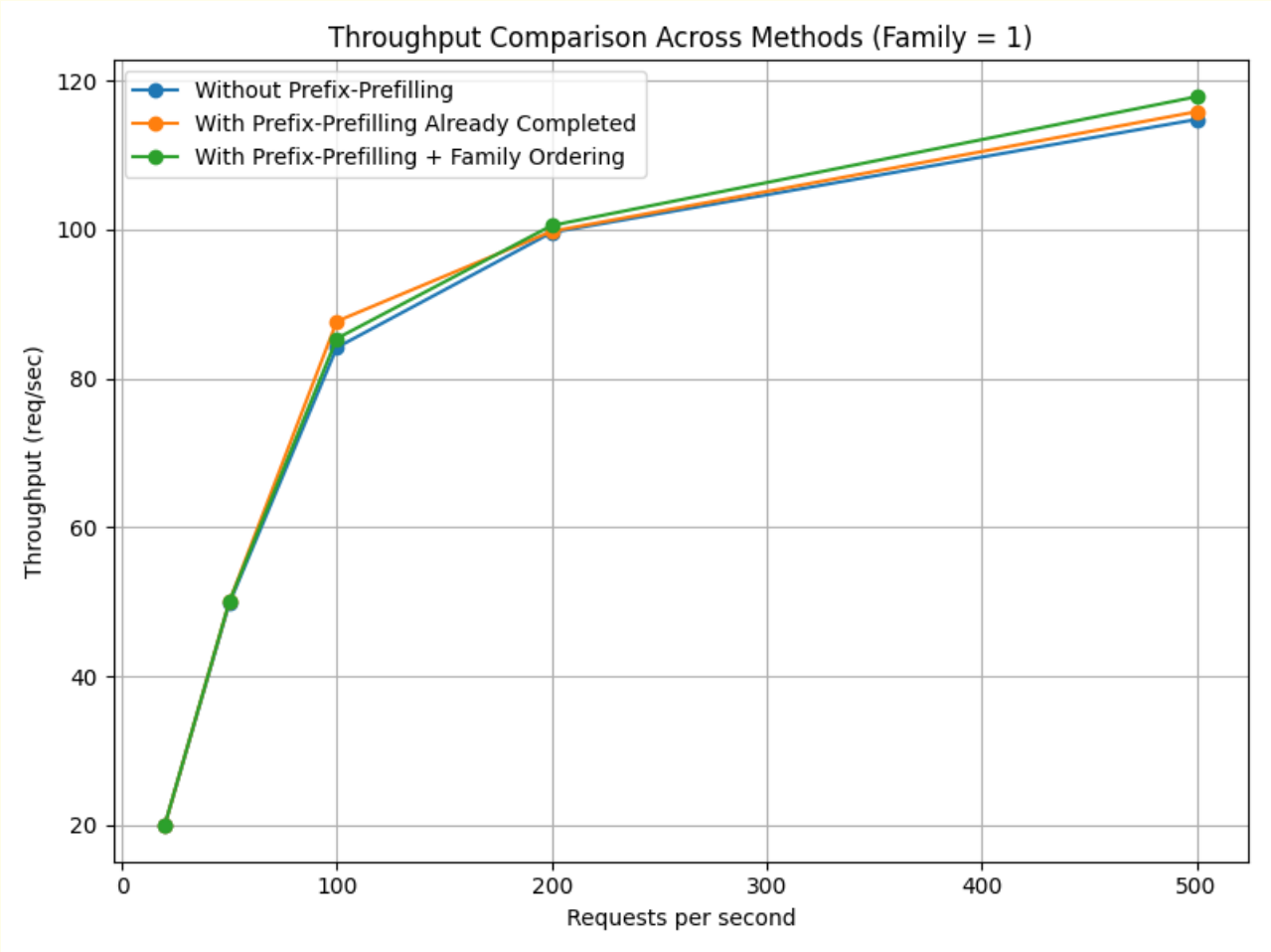


Figure: Throughput vs Load for 1 family

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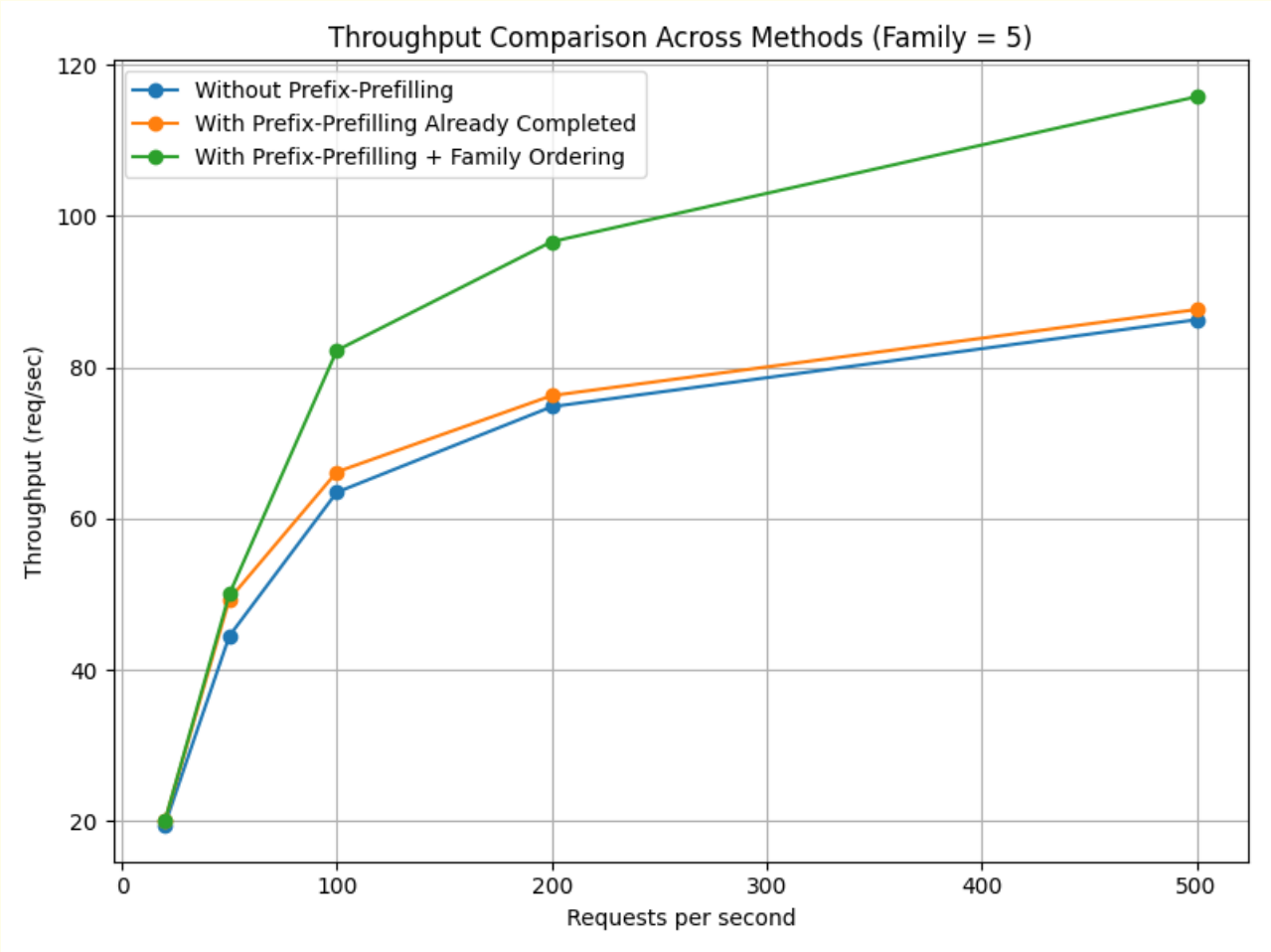


Figure: Throughput vs Load for 5 families

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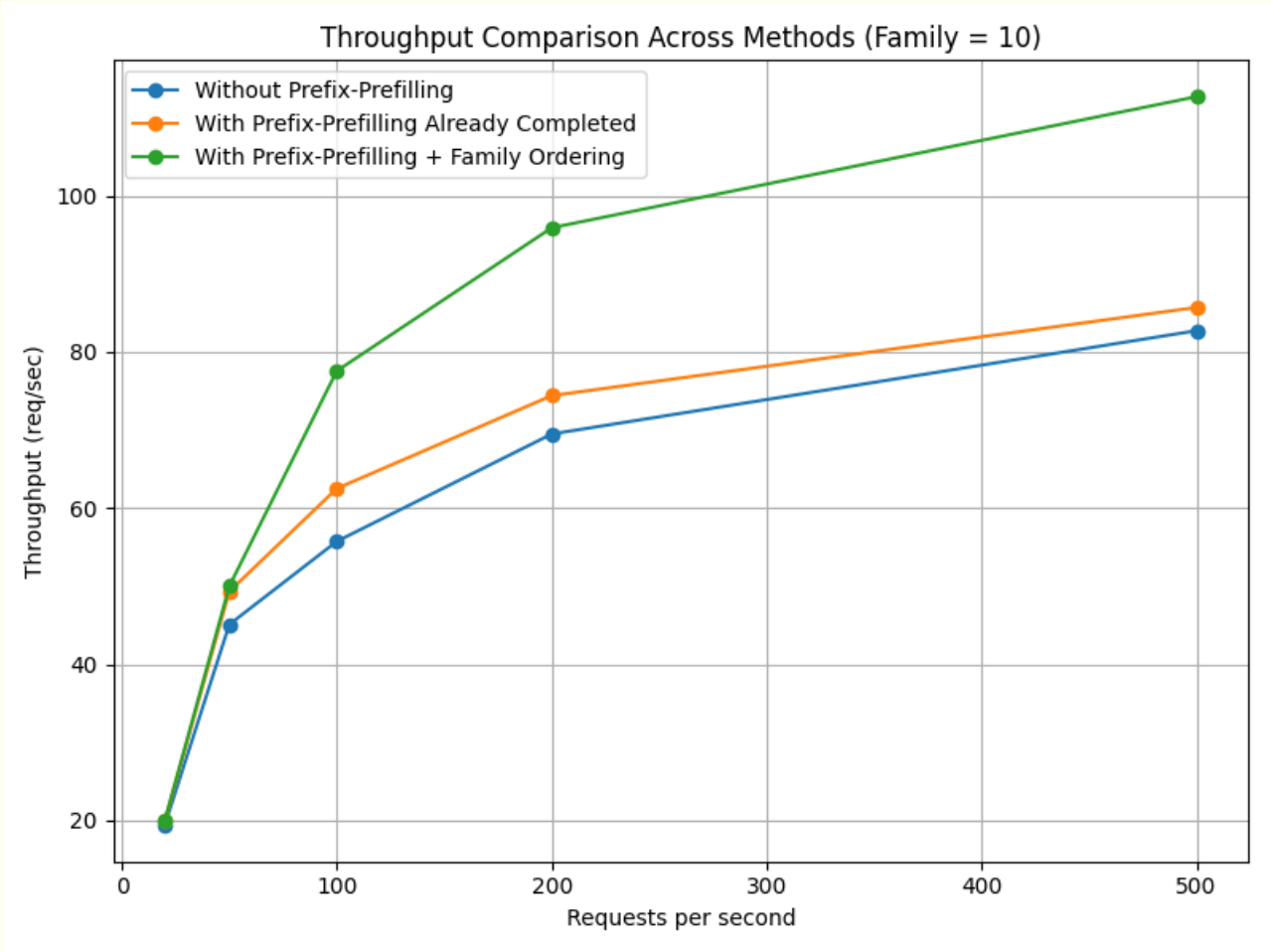


Figure: Throughput vs Load for 10 families

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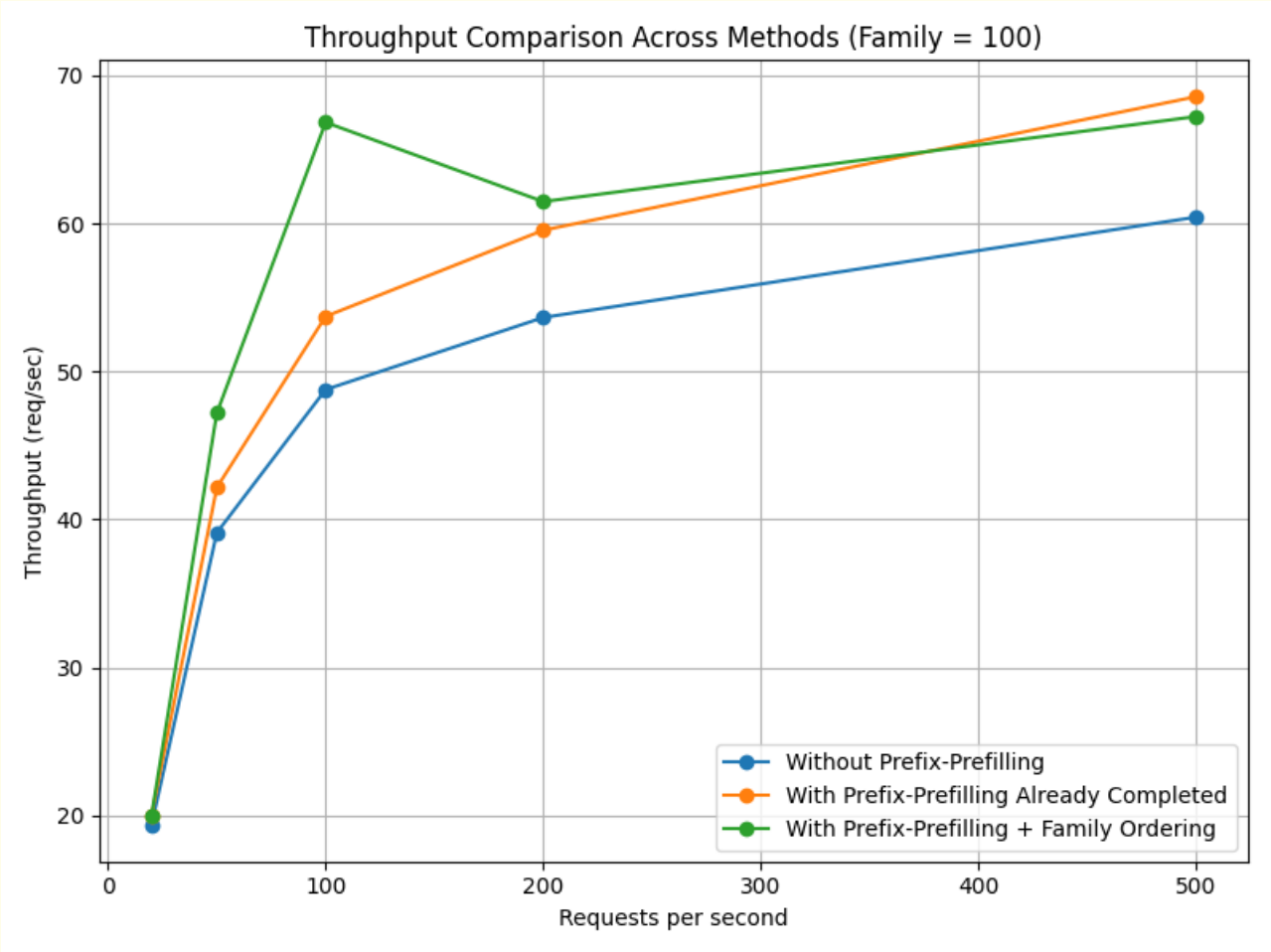


Figure: Throughput vs Load for 100 families

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Broader Goal: Motivation



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- **Without prefix sharing:** batch limited by model context length (e.g., 32,768 tokens for a small Qwen model $\rightarrow$ 1500 tokens $\times$ 20 requests).
- **With prefix sharing:** utilization-limited, not context-limited. As the number of concurrent requests grows, per-request latency increases, and beyond a point, overall throughput would start declining.

Next Step

Design a scheduler for families of such requests. At each scheduling step, we must decide whether to prefill a shared prefix or decode tokens for an existing request.

- Explore KV cache offloading to CPU for extended workloads.

Future Directions: Quality vs Performance



- Strategy 1: Pass full system prompt each time.
- Strategy 2: Fine-tune for families of similar requests.
- Study trade-offs:
  - **Performance:** latency, throughput, memory use.
  - **Quality:** output accuracy, coherence.



Workload Picked

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Thank You